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Medicament preparation devices, methods, and systems

Abstract

A system for preparing a medicament for use by a medicament user includes a proportioning machine with a controller and pumping and clamping actuators to engage a fluid circuit having pumping and clamping portions that engage with respective actuators of the proportioning machine. The fluid circuit includes a mixing container that is initially empty and later filled with a concentrated medicament from a concentrate container. The proportioning machine is configured to receive purified water and to mix it with the concentrated medicament to produce a medicament and to output the medicament to a medicament consumer in such a way that to the medicament consumer the medicament appears to be provided from a bag of medicament.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/162,388 filed Mar. 17, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND

(1) The disclosed subject matter relates generally to devices, methods, systems, improvements, and components for preparing medicaments and making medicament available for use by a consumer, for example, a dialysis cyclor.

(2) Peritoneal dialysis is a mature technology that has been in use for many years. It is one of two common forms of dialysis, the other being hemodialysis, which uses an artificial membrane to directly cleanse the blood of a renal patient. Peritoneal dialysis employs the natural membrane of the peritoneum to permit the removal of excess water and toxins from the blood.

(3) In peritoneal dialysis, sterile peritoneal dialysis fluid is infused into a patient's peritoneal cavity using a catheter that has been inserted through the abdominal wall. The fluid remains in the peritoneal cavity for a dwell period. Osmotic exchange with the patient's blood occurs across the peritoneal membrane, removing urea and other toxins and excess water from the blood. Ions that need to be regulated are also exchanged across the membrane. The removal of excess water results in a higher volume of fluid being removed from the patient than is infused. The net excess is called ultrafiltrate, and the process of removal is called ultrafiltration. After the dwell time, the dialysis fluid is removed from the body cavity through the catheter.

SUMMARY

(4) Methods, device, and systems for preparing medicaments such as, but not limited to, dialysis fluid are disclosed. In embodiments, medicament is prepared at a point of care (POC) automatically using a daily sterile disposable fluid circuit, one or more concentrates to make batches of medicament at the POC. The dialysis fluid may be used at the POC for any type of renal replacement therapy, including at least peritoneal dialysis, hemodialysis, hemofiltration, and hemodiafiltration.

(5) In embodiments, peritoneal dialysis fluid is prepared at a point of use automatically using a daily sterile disposable fluid circuit and one or more long-term concentrate containers that are changed only after multiple days (e.g. weekly). The daily disposable may have concentrate containers that are initially empty and are filled from the long-term concentrate containers once per day at the beginning of a treatment.

(6) Embodiments of medicament preparation, devices, systems, and methods are described herein. The features, in some cases, relate to automated dialysis such as peritoneal dialysis, hemodialysis and others, and in particular to systems, methods, and devices that prepare peritoneal dialysis fluid in a safe and automated way at a point of care. The disclosed features may be applied to any kind of medicament system and are not limited to dialysis fluid.

(7) In embodiments, a system that prepares a medical fluid is configured in such a manner that it outputs the medical fluid to a consuming process (for example, a peritoneal dialysis cyclor) wherein the consuming process does not distinguish between the system that prepares the medical fluid and pre-packaged bags of dialysate. This allows embodiments of the presently disclosed system for preparing the medical fluid to be used with any type of a cyclor, without any special customization or modification of the cyclor.

(8) Objects and advantages of embodiments of the disclosed subject matter will become apparent from the following description when considered in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments will hereinafter be described in detail below with reference to the accompanying drawings, wherein like reference numerals represent like elements. The accompanying drawings have not necessarily been drawn to scale. Where applicable, some features may not be illustrated to assist in the description of underlying features.

(2) FIG. 1A shows a system for preparing a ready to use medicament from concentrated medicament and water according to embodiments of the disclosed subject matter.

(3) FIG. 1B shows another example of a system for preparing a ready to use medicament from concentrated medicament and water according to embodiments of the disclosed subject matter.

- (4) FIG. 1C shows another example of a system for preparing a ready to use medicament from concentrated medicament and water according to embodiments of the disclosed subject matter.
- (5) FIGS. 2A and 2B show a flow chart of a method for preparing a ready-to-use medicament according to embodiments of the disclosed subject matter.
- (6) FIG. 3 shows a system for generating purified water for the system and method of FIGS. 1A and 1B according to embodiments of the disclosed subject matter.
- (7) FIG. 4 shows configurations of the systems providing water to a mixing container according to embodiments of the disclosed subject matter.
- (8) FIG. 5 shows configurations of the systems providing medicament concentrate to the mixing container according to embodiments of the disclosed subject matter.
- (9) FIG. 6 shows configurations of the systems mixing the content of the mixing container according to embodiments of the disclosed subject matter.
- (10) FIG. 7 shows various configurations of the systems testing conductivity of the content of the mixing container according to embodiments of the disclosed subject matter.
- (11) FIG. 8 shows configurations of the systems providing the content of the mixing container to a consumer of the content according to embodiments of the disclosed subject matter.
- (12) FIG. 9 shows a computer system that may describe the functions and elements of a controller as described herein and in accordance with the embodiments of the disclosed subject matter.

DETAILED DESCRIPTION

- (13) FIG. 1A shows an embodiment of a system that uses water and concentrated medicament **199** (also referred to as “medicament concentrate” or “concentrate”) to make a therapeutic fluid that can be used for treatment according to embodiments of the disclosed subject matter. In embodiments, the concentrated medicament may be a dextrose solution. Referring to FIG. 1A, a purified water source **133** with a water pump **113** supplies highly purified water through a connector **124** through a water line **142**. The water line **142** has a non-reopenable clamp **146**, another connector **124**, a manual tube clamp **101**, and a pair of redundant 0.2 micron sterilizing filters **112**, as shown. In embodiments, different types of sterilizing filters may be used, and not limited to 0.2 micron, or to two redundant filters. For example, a single filter may be used and a testing protocol provided to ensure that the filter does not fail before replacement.
- (14) A water inlet clamp **138**, batch release clamp **136**, and a conductivity sensor clamp **140** are controlled by a controller **141**, which may be operatively coupled to a user interface **143**, which may include a visual and/or audible output and various devices for receiving user input. The controller **141** controls the pinch clamps and a peristaltic pump **129** to make a batch of diluted concentrate in a mixing container **102** by diluting medicament concentrate (e.g., dialysis fluid concentrate) in the mixing container **102**. The mixing container **102** is supplied empty and permanently connected to a fluid circuit that includes fluid lines **149**, **123**, and **125**.
- (15) A pressure sensor **301** is provided in the flow path as shown and outputs a signal representative of the pressure in the fluid lines that are fluidly connected to the pressure sensor. This pressure signal may be provided to controller **141**.
- (16) In embodiments, the medicament concentrate **199** is provided in a separate package that is connected via connector **124** to concentrate line **249** as shown. The concentrate line **249** may include an optional filter **122**. The filter **122** may be a touch contamination protection filter, such as a 0.2 micron filter.
- (17) The mixing container **102** may be a part of a disposable component **161** that is replaced regularly, such as with each batch, every day, every week, or every month. In an embodiment, the mixing container **102** is empty initially when the disposable component **161** is connected to the system.
- (18) The mixing container **102** may be made of a flexible material, such as a polymer so its shape is not rigid. To provide support for the mixing container **102**, it is held by a tub **106** which is sufficiently rigid to support the mixing container **102** when it is full of fluid. A leak sensor **107** is

provided in the tub **106** and it detects leaks into the tub **106** while a temperature sensor **109** may also be provided in or on the tub **106** and it detects the temperature of the fluid in the mixing container **102**. A warmer **104** may be provided as shown to provide heat to tub **106**, but the warmer **104** may be omitted if another heater exists elsewhere in the system. Note that the concentrate **199** that will be supplied to the mixing container **102** may be used for making any type of medicament, not just dialysis fluid.

(19) To supply water to mixing container **102**, clamp **139** can remain closed, and pump **129** runs to move the water from water line **142** to supply line **123** and mixing container **102** while valve **138** is open. Also, to make the medicament available to the medicament user **157**, clamps **136** and **139** are opened and the other clamps are closed. There is no backpressure provided by a cracking check valve as in the embodiment of FIG. **1A**. Thus, the medicament pump **115** may draw from the mixing container **102** without the assistance of a predefined backpressure, hence without the use of peristaltic pump **129**. Alternatively, the peristaltic pump **129** may be run through a circulating path of **149**, **123**, and **125** with a feedback-controlled clamp **139** according to pressure indicated by pressure sensor **301**. Here, clamps are closed except for **136** and **139** and the medicament user draws from a pressurized line.

(20) FIG. **1B** shows a medicament generation system that is like that of FIG. **1A** except that there is no valve **139** and instead cracking pressure check valve **154** is provided. The check valve **154** prevents flow in line **125** out of mixing container **102**, and allows flow into mixing container **102** only when the cracking pressure is overcome. The cracking pressure may be selected at 3.5 PSI in embodiments.

(21) Likewise, a check valve **151** may be added to the concentrate supply line **249** as shown, preventing back flow of concentrate into the container of concentrate **199**. In embodiments, this allows for the safe preparation of multiple batches of diluted medicament from the same container of concentrate **199**, as back flow (which is undesirable) into the concentrate container is prevented. In addition, the concentrate fill line **249** is routed to an opposite side of the peristaltic pump **129** as compared with FIG. **1A**. In embodiments, the cracking pressure of check valve **151** is lower than the cracking pressure of check valve **154**. When the peristaltic pump **129** rotates in a direction that draws concentrate from the container of concentrate **199**, the difference in the cracking pressures helps ensure that contents of the mixing container **102** are not drawn out of the container when concentrate is being drawn through concentrate line **249**.

(22) Another difference with respect to FIG. **1A** is the presence of two conductivity/temperature sensors **159c** and **159s**, but it will be understood that the single conductivity/temperature sensor **159** in FIG. **1A** can be replaced with two sensors as shown in FIG. **1B**.

(23) Note that in variations of most of the embodiments, the purified water source **133** may include a container or containers of purified water such as one or more polymer bags. In such embodiments, there may be a water pump arranged in a “pull” configuration. In any of the embodiments, the medicament user **157** may include a pump, such as the pump **115** illustrated in FIG. **1A**. For example, the medicament user **157** may include a dialysis cycler that is configured to draw from a container of dialysis fluid.

(24) To permit the medicament user **157** to draw medicament on-demand, the controller **141** may be programmed to maintain a constant pressure that is compatible with a pump in the medicament user **157**. For example, the pressure-based control using the pressure sensor **301** may maintain a pressure that mimics a simple container that allows the medicament user **157** to draw from a container of dialysis fluid.

(25) In embodiments, the medicament user **157** can use its own pump, such as the pump **115**, to move fluid from the mixing container **102** without the use of pump **129**. In this example, valves **136** and **139** will be opened, and the medicament user **157** will operate its pump to draw fluid from the mixing container **102**.

(26) FIG. **1C** shows another variation of a medicament generation system of FIG. **1B**, except that

cracking check valve **154** is replaced with valve **139**. Similar to FIG. **1B**, a check valve **151** may be added to the concentrate supply line **249** as shown, preventing back flow of concentrate into the container of concentrate **199**. In embodiments, this allows for the safe preparation of multiple batches of diluted medicament from the same container of concentrate **199**, as back flow (which is undesirable) into the concentrate container is prevented.

(27) FIG. **2A** shows a procedure for reliably measuring the conductivity of a fluid. The fluid circuit will be configured as shown in FIG. **7**. In this procedure two consecutive measurements are made of conductivity and temperature at different times so that the conductivity is measured for two different parts of a flow stream. The two consecutive measurements can be made with a single sensor **159** at two different times, or they may be made using two different sensors such as **159c** and **159s**. If the two different readings are within a predefined range of each other, the controller **141** mixes the mixing container **102** a second time. The measurements are compared again and if the two conductivity are within a predefined range of each other, the measurement is output as correct. If the two measurements show a difference in concentration beyond the predefined range, then the mixing container is mixed again (configuration of FIG. **6**) and two consecutive measurements are taken again. The contents of drain line **147** may be purged to the drain. The rationale behind this is that a difference in magnitude of the consecutive measurements may be caused by inadequate mixing. If, after mixing again and repeating the two consecutive measurements, the magnitudes are still outside of the predefined range of each other, then the controller outputs a measurement failure or data indicating “no measurement.” Also, after the initial measurement the controller determines if there is gross disparity between the measurement and a predefined or calculated estimate then the algorithm will immediately output an indication and stop the process.

(28) Referring to FIG. **2A**, at **51**, the fluid whose conductivity is to be measured is pumped through conductivity/temperature sensors **159c** and **159s** by opening the conductivity sensor clamp **140** and closing the others, as shown in FIG. **7**. At **S3**, the peristaltic pump **129** is run in a direction indicated by the arrows as shown in FIG. **7**. The conductivity is measured a first time by flowing mixed fluid from the mixing container **102** through the temperature and conductivity sensors **159c** and **159s** (or single conductivity sensor **159**, depending on the configuration of the system) and storing a magnitude or multiple magnitude readings thereof. If the absolute value of the difference between the measured conductivity readings is greater than a predefined magnitude at **S5**, then control goes to **S27** where an error indication is output. Otherwise, at **S7**, additional fluid is pumped from the mixing container **102** and at **S9**, the conductivity is measured a second time at **S9**. At **S11** it is determined if the first and second measurements agree within a predefined range. If the measurements differ by less a than predefined range, then the measurement is output at **S13** where the output measurement may be one of the first and second measurements or an average of the measured values. If the measurements differ by more than the predefined range, then control proceeds to **S15** where the mixing container contents are mixed again (because it is assumed that the measurements may differ due to insufficient mixing such that the medicament is not yet uniformly mixed in the mixing container **102**). At **S17**, a third measurement for the conductivity is obtained. If the measured conductivity differs from the expected conductivity by a predefined magnitude at **5171**, a gross error is detected at **S27**. Otherwise, the process continues at **S19**, where the mixing container contents are again pumped through the conductivity sensors **159c** and **159s** and a fourth measurement of conductivity is made at **S21** in the manner described above. At **S23** it is determined if the third and fourth measurement are within the predefined range and if so, at **S25**, the measured values (average of the two sensors or one of them) are output at **S13** as a valid conductivity measurement. If the measured values still disagree by the predefined amount, then at **S25** a failure is output.

(29) Note that the consecutive measurements may be done sequentially in time using one temperature-compensated conductivity measurement indicated by conductivity/temperature sensor

159c, only. The fluid then is conveyed, and a temperature-compensated conductivity measurement is measured again by the same sensor. In alternative embodiments, separate pairs or single temperature-compensating may be separated along a line and the measurement generated by them may be compared instead.

(30) FIG. 2B shows a flow chart for a procedure that may be executed by the controller **141** with respect to the embodiment of FIG. 1A, 1B, or 1C. It incorporates the procedure of FIG. 2A by the reference to “conductivity test” described with reference to the procedure of FIG. 2A. When the conductivity test is referenced it means the procedure of FIG. 2A is entered and upon exiting proceeds to the next procedure element in FIG. 2B.

(31) At **S10**, water is added by pumping it into the mixing container **102** from the purified water source **133**. This is done by placing the system in the configuration of FIG. 4. The water pump **113** and the peristaltic pump **129** are activated for a predefined number of cycles or a predefined time interval, resulting in a quantify of water being conveyed along water line **142**, through opened valve **138**, through transfer line **149**, through peristaltic pump **129** and through connector line **123** into mixing container **102**.

(32) As shown in FIG. 4, valve **139** can be opened so that the water pump **113** alone, without the involvement of peristaltic pump **129**, conveys water into the mixing container **102** through line **125**. Alternatively, valve **139** can be closed and peristaltic pump **129** operates to move water from water supply line **149** to inlet line **123** and through the inlet line **123** into mixing container **102**.

(33) It will be understood that the two pumps **113** and **129** are controlled such that the water pressure in the line is below the cracking pressure of the check valve **154** in the embodiment of FIG. 1B. This way, the water enters the mixing container only through supply line **123**. On the other hand, in the embodiment of FIGS. 1A and 1C, the additional valve **139** can be closed to ensure that water does not flow through supply line **125**. Note that valve **139** is not present in the embodiment of FIG. 1B. Further, the pumps are controlled to hold a steady pressure to provide a consistent upstream pressure for the peristaltic pump **129**.

(34) The amount of fluid conveyed at **S10** may be a fraction of the total estimate required for a sufficient level of dilution, such as 50% of the expected total water volume.

(35) Next, a quantity of concentrated medicament **199** is conveyed from the medicament container through medicament supply line **249**, past the valve **144** into supply line **149** and through the supply line **129** into mixing container **102**, as shown in the configuration of FIG. 5. It is noted that in this configuration, valve **144** is opened while valve **139** is closed. At this stage in the process a quantity of concentrated medicament and water is present in mixing container **102**. As noted above, the quantity of water that is present may be smaller than the final quantity of water that is expected to be needed to completely dilute the concentrated medicament into its final concentration. Then, the contents of the mixing container **102** are mixed as shown in the configuration of the system in FIG. 6.

(36) Next, at **S16**, the conductivity of the mixing container contents is measured by performing a conductivity test, described in FIG. 2A, as all of the medicament concentrate is already present in the mixing container **102**, so the only possible action is the addition of water. To avoid over-dilution, water is added incrementally, and the conductivity is checked to reduce the possibility of over-dilution.

(37) At **S18**, the controller determines whether the first measurement indicates a gross error by comparing the measured value of conductivity to a fixed predefined range of magnitude representing reasonable conductivities. If the measured value is outside the range, a gross error signal is output, and the batch is failed at **S40**. If not, the control proceeds to **S22** where the additional water, based on the correctly measured value, is calculated. The calculation may be based on a dilution formula or a look-up table, among other options. A fraction of this calculated amount is added at **S24**. The addition of only a fraction at this stage provides a further margin of error, in case there is inaccuracy in the measurement of the water being added (e.g., due to

inaccuracy of a peristaltic pump). Then at **S28**, the conductivity test is performed again. If the measurement is valid at **S30**, then a final fraction of water is calculated at **S32** and added to the mixing container. The calculation of the final amount of water can take into account the expected conductivity at this stage and the measured conductivity, as a reflection of the accuracy of the metering of water, and this can be used to more finely calibrate the pump(s) that supply water, to provide a correct final concentration of medicament. A conductivity test is again performed at **S38**. If the measurement is deemed correct at **S42**, then the medicament is made available for use at **S44**. If not, then the batch is failed at **S40**. A failed batch may result in a message or alert output via the user interface **143**. In embodiments, the failed batch may be drained from the system through drain line **147**. In embodiments, one or more samples of the failed batch may be stored in testing containers in the system (not illustrated) for later analysis and troubleshooting of the system.

(38) Note there may be a single conductivity/temperature sensor, or a pair of conductivity/temperature sensors as shown. A pair of conductivity/temperature sensors may provide a check against poor accuracy or failure of one of the sensors. The fluid from the mixing container flows through the drain conductivity line **147** using the peristaltic pump **129**.

(39) FIG. 3 shows a water treatment plant **200** that may constitute an embodiment the purified water source **133**. The water treatment plant **200** has an initial pretreatment stage that includes a connector **250** to connect to an unfiltered water source **256**, for example a water tap. The water flows through a check valve **150**, through a pressure regulator **254**, and then through a sediment filter **202**. The check valve **150** prevents backflow of the water. The water then flows through an air vent **204** that removes air from the water. The water then flows through a connector **205** that connects to a water shutoff clamp **206**, a snubber **207**, and a water inlet pressure sensor **208**. Water is pumped by water pump **212** which has an encoder **213** for precise tracking of the water pump **212** speed. The snubber **207** reduces pressure fluctuations. The water then flows through a water output pressure sensor **214**, through an ultraviolet light lamp **220** and into a filter plant **337** that performs deionization, carbon filtration, and sterilizing filtration. A UV light sensor **216** may be provided to detect whether the ultraviolet light lamp **220** is operating, so that it can be replaced if it becomes inoperable. A first-use-fuse **218** together with a connector **219** is provided on the inlet of sterilizing filter plant **337**, such that the fuse indicates whether the filter plant **337** has been used. This helps reduce the likelihood that a previously-used filter plant is reused unintentionally. A combined control unit and leak sensor are indicated at **210**. In the sterilizing filter plant **337**, the water flows through a carbon filter **228** and three separated bed deionization filters **226** which may be resin separated bed filters. The water then flows through a mixed bed deionization filter **223** which follows the separated bed filters **226**. The mixed bed deionization filter **223** may be a resin mixed bed filter. The water then flows through first and second ultrafilters **230**, which follow the mixed bed deionization filter **223**, and into the consumer of pure water **234**. The embodiments of FIGS. 1A-1C are examples of a consumer of pure water **234**.

(40) Between a last separated bed deionization filter **226** and the mixed bed deionization filter **223** is a resistivity sensor **222** which indicates when the separated bed deionization filters **226** are nearing exhaustion, or at exhaustion. The mixed bed deionization filter **223** is still able to hold a predefined minimum magnitude of resistivity but the separated bed deionization filters **226** and the mixed bed deionization filter **223** may be replaced at the same time. In embodiments, along with the separated bed deionization filters **226** and the mixed bed deionization filter **223**, the carbon filter **228** and ultrafilters **230** along with the interconnecting lines and other components may also be replaced as a single package. A current treatment can be completed in reliance on the mixed bed deionization filter **223** before the exhausted filters are replaced. A further resistivity sensor **225** detects unexpected problems with the mixed bed deionization filter **223** upstream from the separated bed deionization filters **226**, which may require shutdown of the treatment and immediate replacement of the filters. Note that each of the ultrafilters **230** has an air vent **232**. A check valve **150** is located downstream of the ultrafilters **230**. The consumer of pure water **234** may be unit

such as that of FIGS. 1A-1C which mixes a batch of medicament for use by a medicament user 157 such as a peritoneal dialysis cyclor or any other type of medicament consuming device.

(41) It should be evident from the above that the procedures of FIG. 2B in combination with those of FIG. 2A may be performed using the embodiments of FIGS. 1A-1C.

(42) Note in any of the embodiments where the term clamp is used, it should be recognized that the functional element includes a tube or other flexible conduit and the clamp so that it functions as a valve. In any of the embodiments, another type of valve may be substituted for the clamp and conduit to provide the same function. Such a variation may be considered to alternative embodiments and clamp and conduit are not limiting of the subject matter conveyed herein.

(43) Note that in any of the embodiments that identify a bag as the container, any bag may be replaced by any container including those of glass, polymer and other materials. In any embodiment where flow control is performed by a clamp, it should be understood that in any embodiment, including the claims, any clamp can be replaced by another type of valve such as a stopcock valve, a volcano valve, a ball valve, a gate valve or other type of flow controller. It should be understood that a clamp in the context of the disclosed subject matter is a clamp that closes around a tube to selectively control flow through the position of the clamp. Note that in any of the embodiments, the order of adding and mixing to the mixing container 102 can be reversed from what is described with respect to the embodiments. In any of the embodiments instead of dextrose concentrate being used, this can be substituted for glucose or another osmotic agent.

(44) FIG. 4 shows a first step that adds water to the mixing container 102 from the water source 133. The peristaltic pump 129 runs in a direction to pump water through the first mixing container connector line 123 and all clamps are closed except for clamp 138. Optionally, clamp 139 may be opened, as shown.

(45) Referring to FIG. 4, water is provided from the purified water source 133 to the system. The peristaltic pump 129 is configured to move fluid in a line 123 connected to the mixing container 102. The peristaltic pump 129 also moves fluid, at selected times, through the line 125 which returns the fluid to the mixing bag. Line 125 can be provided with a check valve 154 (FIG. 1B) which prevents flow in one direction and has a cracking pressure which must be overcome for water to flow in the other direction. In the example of FIG. 1B, the check valve permits water to flow through line 125 toward the mixing container 102 when the cracking pressure of the check valve 154 is overcome. Initially, the purified water from the purified water source 133 is pumped by the water pump 113 with water inlet clamp 138 open and the batch release clamp 136 and the conductivity sensor clamp 140 closed such that water is pumped into the mixing container 102 through line 123 with the peristaltic pump 129 running so as to convey water into the mixing container 102, as shown in FIG. 4.

(46) Still Referring to FIG. 4, the peristaltic pump 129 may remain turned off while clamp 139 is opened, thereby allowing pressure generated by pump 113 to convey the purified water through line 125 into mixing container 102.

(47) FIG. 5 illustrates the configuration of the system when medicament concentrate 199 flows into the mixing container 102. As shown in the figure, valve 144 is opened and peristaltic pump 129 can operate in reverse direction relative to when it is used to fill the mixing container 102 with water, such that the concentrate flows through inlet line 123 into mixing container 102. In an alternate embodiment, the medicament concentrate 199 can be positioned sufficiently high or above mixing container 102 that a gravity powered fill can be accomplished. In this scenario, valve 144 is opened and valve 139 is opened which permits gravity to convey the medicament concentrate through inlet line 125 into mixing container 102.

(48) Referring to FIG. 6, to mix the contents of the mixing container 102 the peristaltic pump 129 pumps fluid in a circular path through lines 123 and 125 with all the clamps closed except for clamp 139. Then the contents of the mixing container 102 are mixed by the flow circulating through the mixing container 102. It will be noted that because there is no check valve on line 125

in this embodiment, the peristaltic pump **129** does not have to generate pressure which is sufficient to overcome the cracking pressure of the check valve **154** that is shown in FIG. **1B**.

(49) Referring to FIG. **7**, after a sufficient time of mixing, a sample of the fluid in the mixing container **102** may be pumped through a drain conductivity line **147** which contains conductivity/temperature sensors **159c** and **159s** (control sensor **159c** and safety sensor **159s**) to determine a temperature-compensated conductivity of the diluted medicament. Each sensor **159c** and **159s** may be configured to calculate conductivity and temperature of fluid passing through or past the sensor. Valve **140** is opened and the peristaltic pump **129** operates in reverse direction as shown in the figure. Two redundant sensors **159c** and **159s** may be provided, to enable a comparison of their respective measurements and thereby to confirm that the sensors are functioning. If their respective measurements are within a predetermined range, the sensors are understood to be functioning correctly. On the other hand, if their respective measurements are outside of the predetermined range, an error condition may be signaled as described below.

(50) Referring now to FIG. **8**, once of the medicament is prepared and mixed in the mixing container at **102**, and the medicament is deemed to be ready for use, the batch release clamp **136** is open and the water inlet clamp **138** and the conductivity sensor clamp **140** are closed. A pump **115** in a medicament user **157** may then draw fluid from the circular path as the peristaltic pump **129** rotates to maintain fluid at the cracking pressure of the check valve **154** in FIG. **1B**, or at a pressure that is controlled based on a pressure signal from pressure sensor **301**, if no cracking check valve is used (e.g., FIG. **1A**, **1C**). At this time, the water inlet clamp **138** and the conductivity sensor clamp **140** are closed. The medicament user **157** may be any type of treatment device or container that receives the mixed medicament from the mixing container **102**. In embodiments, the cracking pressure may be 3.5 PSI. It will be understood that this makes the medicament preparation system appear like a bag of dialysate with a head pressure of 3.5 PSI.

(51) A medicament pump **115** in the medicament user **157** may see a positive pressure at the cracking pressure type check valve **154** cracking pressure, which may facilitate the pump **115** of the medicament user **157** by mimicking the pressure of an elevated medicament container with a head pressure approximately at the cracking pressure of the check valve **154**. In embodiments, clamp **139** is closed while peristaltic pump **129** operates in the direction shown in the drawing. Clamp **136** is opened and the medicament is conveyed through supply lines **137** and **153** to medicament user **157**. A pressure sensor **301** is provided to measure the pressure in this fluid channel and to provide a signal, which may be used in feedback control, to modulate the speed of the peristaltic pump **129** and thereby provide a predetermined pressure in the formed fluid channel. In further embodiments, the peristaltic pump **129** is not used, and instead medicament user pump **115** operates to draw the medicament from the mixing container **102**. Clamp **139** and clamp **136** are both opened, thereby providing a fluid path between the mixing container **102** and the medicament user **157**. It is possible to elevate mixing container **102** to such a level that it provides a positive pressure (head pressure) for the medicament user **157**.

(52) Note that temperature-compensated conductivity is intended to refer to a number that is proportional to concentration and may be determined in various ways including but not limited to a lookup table and a formula. For the remainder of this disclosure, a reference conductivity may be understood to mean temperature-compensated conductivity or an actual calculation of concentration. That is, the temperature-compensated conductivity may be a value that is generated by the controller by multiplying the measured conductivity with a value that represents the rate of change of concentration with temperature. In other embodiments, the controller **141** may calculate a concentration directly using a look-up table or formula.

(53) As noted above, the mixing container at **102** may be part of a disposable unit **161**. Included in a disposable unit **161** are a source medicament supply line **137**, transfer line **149**, water source line **142**, drain conductivity line **147** and the mixing container **102**. The disposable unit **161** is permanently interconnected up to and including an end of each of the connectors **124**. Also

included in the disposable unit **161** may be the check valve **154** that has a predefined cracking pressure (e.g., 3.5 PSI). The disposable unit **161** can be connected to check valve **150** which prevents back flow in the drain conductivity line **147**. Mixed fluid is pumped through temperature and conductivity sensors **159c** and **159s** and is determined to be mixed when two consecutive measurements of the conductivity of mixed fluid flowing through the temperature and conductivity sensors **159c** and **159s** are within a predefined range of each other. If two consecutive measurements of the conductivity differ by a margin greater than the predefined range, the mixing container **102** may be mixed again. An attachment to drain or waste container is provided by a connector **152**. Note the mixing bag may contain a liquid or dry concentrate which forms part of the disposable unit **161**.

(54) A door lock **116** is provided adjacent a user interface door **105** to lock the user interface door. A physical door **105** that opens encloses and provides access to the interior of the fluid preparation system may have a user interface on it which may be a part of user interface **143**. A door sensor **118** detects whether the door lock is in an open or a locked position to ensure that all clamps and the peristaltic pump actuators are fully engaged with the disposable fluid circuit.

(55) The door sensor **118** may include a plunger which is pressed in when the door is closed and outputs an electrical signal to indicate whether or not the door is closed. In other embodiments, the door sensor **118** may include a magnetic reed switch which detects the presence or the absence of a magnet which is located on the door **105** at a location which is detectable by the reed switch. Purified water flows into the disposable circuit where a pair of 0.2 micron filters (also in the disposable unit **161**) are located to ensure that any touch contamination is prevented from flowing into the disposable circuit. An optional sterilizing filter **120** may be provided in a user medicament supply line **153**. The mixing container **102** of the disposable unit **161** may have sufficient volume for a single treatment or in embodiments, multiple treatments. To make a batch of dilute concentrate, water is pumped into the mixing container **102** which contains concentrate sealed in it as-delivered.

(56) The medicament output line **137** may include an optional air removal filter **121**. The air removal filter **121** may be a 1.2 μm filter which removes air.

(57) The check valve **150** in drain conductivity line **147** ensures the flow does not reverse to safeguard against contamination in the medicament or water lines or other sterile fluid circuits. Note that the peristaltic pump **129** is regulated to ensure the output pressure remains below the cracking pressure of the check valve **154** when the conductivity of the mixing container contents is measured.

(58) FIG. 9 shows a block diagram of an example computer system according to embodiments of the disclosed subject matter. In various embodiments, all, or parts of system **1000** may be included in a medical treatment device/system such as a renal replacement therapy system. In these embodiments, all, or parts of system **1000** may provide the functionality of a controller of the medical treatment device/systems. In some embodiments, all, or parts of system **1000** may be implemented as a distributed system, for example, as a cloud-based system.

(59) System **1000** includes a computer **1002** such as a personal computer or workstation or other such computing system that includes a processor **1006**. However, alternative embodiments may implement more than one processor and/or one or more microprocessors, microcontroller devices, or control logic including integrated circuits such as ASIC.

(60) Computer **1002** further includes a bus **1004** that provides communication functionality among various modules of computer **1002**. For example, bus **1004** may allow for communicating information/data between processor **1006** and a memory **1008** of computer **1002** so that processor **1006** may retrieve stored data from memory **1008** and/or execute instructions stored on memory **1008**. In one embodiment, such instructions may be compiled from source code/objects provided in accordance with a programming language such as Java, C++, C#, .net, Visual Basic™ language, LabVIEW, or another structured or object-oriented programming language. In one embodiment, the

instructions include software modules that, when executed by processor **1006**, provide renal replacement therapy functionality according to any of the embodiments disclosed herein.

(61) Memory **1008** may include any volatile or non-volatile computer-readable memory that can be read by computer **1002**. For example, memory **1008** may include a non-transitory computer-readable medium such as ROM, PROM, EEPROM, RAM, flash memory, disk drive, etc. Memory **1008** may be a removable or non-removable medium.

(62) Bus **1004** may further allow for communication between computer **1002** and a display **1018**, a keyboard **1020**, a mouse **1022**, and a speaker **1024**, each providing respective functionality in accordance with various embodiments disclosed herein, for example, for configuring a treatment for a patient and monitoring a patient during a treatment.

(63) Computer **1002** may also implement a communication interface **1010** to communicate with a network **1012** to provide any functionality disclosed herein, for example, for alerting a healthcare professional and/or receiving instructions from a healthcare professional, reporting patient/device conditions in a distributed system for training a machine learning algorithm, logging data to a remote repository, etc. Communication interface **1010** may be any such interface known in the art to provide wireless and/or wired communication, such as a network card or a modem.

(64) Bus **1004** may further allow for communication with one or more sensors **1014** and one or more actuators **1016**, each providing respective functionality in accordance with various embodiments disclosed herein, for example, for measuring signals.

(65) It will be appreciated that the modules, processes, systems, and sections described above can be implemented in hardware, hardware programmed by software, software instruction stored on a non-transitory computer readable medium or a combination of the above. For example, a method for providing a medicament to a medicament user can be implemented, for example, using a processor configured to execute a sequence of programmed instructions stored on a non-transitory computer readable medium. For example, the processor can include, but not be limited to, a personal computer or workstation or other such computing system that includes a processor, microprocessor, microcontroller device, or is comprised of control logic including integrated circuits such as, for example, an Application Specific Integrated Circuit (ASIC). The instructions can be compiled from source code instructions provided in accordance with a programming language such as Java, C++, C#.net or the like. The instructions can also comprise code and data objects provided in accordance with, for example, the Visual Basic™ language, LabVIEW, or another structured or object-oriented programming language. The sequence of programmed instructions and data associated therewith can be stored in a non-transitory computer-readable medium such as a computer memory or storage device which may be any suitable memory apparatus, such as, but not limited to read-only memory (ROM), programmable read-only memory (PROM), electrically erasable programmable read-only memory (EEPROM), random-access memory (RAM), flash memory, disk drive and the like.

(66) Furthermore, the modules, processes, systems, and sections can be implemented as a single processor or as a distributed processor. Further, it should be appreciated that the steps mentioned above may be performed on a single or distributed processor (single and/or multi-core). Also, the processes, modules, and sub-modules described in the various figures of and for embodiments above may be distributed across multiple computers or systems or may be co-located in a single processor or system. Exemplary structural embodiment alternatives suitable for implementing the modules, sections, systems, means, or processes described herein are provided below.

(67) The modules, processors or systems described above can be implemented as a programmed general purpose computer, an electronic device programmed with microcode, a hard-wired analog logic circuit, software stored on a computer-readable medium or signal, an optical computing device, a networked system of electronic and/or optical devices, a special purpose computing device, an integrated circuit device, a semiconductor chip, and a software module or object stored on a computer-readable medium or signal, for example.

(68) Embodiments of the method and system (or their sub-components or modules), may be implemented on a general-purpose computer, a special-purpose computer, a programmed microprocessor or microcontroller and peripheral integrated circuit element, an ASIC or other integrated circuit, a digital signal processor, a hardwired electronic or logic circuit such as a discrete element circuit, a programmed logic circuit such as a programmable logic device (PLD), programmable logic array (PLA), field-programmable gate array (FPGA), programmable array logic (PAL) device, or the like. In general, any process capable of implementing the functions or steps described herein can be used to implement embodiments of the method, system, or a computer program product (software program stored on a non-transitory computer readable medium).

(69) Furthermore, embodiments of the disclosed method, system, and computer program product may be readily implemented, fully or partially, in software using, for example, object or object-oriented software development environments that provide portable source code that can be used on a variety of computer platforms. Alternatively, embodiments of the disclosed method, system, and computer program product can be implemented partially or fully in hardware using, for example, standard logic circuits or a very-large-scale integration (VLSI) design. Other hardware or software can be used to implement embodiments depending on the speed and/or efficiency requirements of the systems, the particular function, and/or particular software or hardware system, microprocessor, or microcomputer being utilized. Embodiments of the method, system, and computer program product can be implemented in hardware and/or software using any known or later developed systems or structures, devices and/or software by those of ordinary skill in the applicable art from the function description provided herein and with a general basic knowledge of control systems of medical devices and/or computer programming arts.

(70) Moreover, embodiments of the disclosed method, system, and computer program product can be implemented in software executed on a programmed general-purpose computer, a special purpose computer, a microprocessor, or the like.

(71) According to a first further embodiment, there is provided a system for preparing a medicament for use by a medicament user, including: a proportioning machine with a controller (**141**, **1002**) and pumping and clamping actuators (**1016**) to engage a fluid circuit having pumping and clamping portions that engage with respective actuators among the pumping and clamping actuators (**1016**); the fluid circuit having an empty mixing container (**102**) attached to the fluid circuit; a detachable container (**199**) having concentrated medicament therein; the proportioning machine being configured to flow fluid from the mixing container (**102**) into and out of the mixing container (**102**) to circulate the fluid; the proportioning machine being configured to flow water and the concentrated medicament into said mixing container (**102**) to dilute the concentrated medicament to make a ready-to-use medicament; the proportioning machine controller (**141**) being configured to regulate a clamp (**139**) on a return line (**125**) leading to said mixing container (**102**) to generate a predefined pressure in an outlet line (**137**, **153**) of the fluid circuit which is attachable to an external user (**157**) of the ready-to-use medicament; and the predefined pressure being maintained in the outlet line (**137**, **153**) by pressure feedback control.

(72) According to a second further embodiment, there is provided the system of the first further embodiment, wherein the clamp (**139**) is a controllable clamp that regulates flow and pressure in a line (**125**). According to a third further embodiment, there is provided the system of the first further embodiment or any of the other foregoing embodiments, wherein the concentrate and ready-to-use medicament are for peritoneal dialysis fluid. According to a fourth further embodiment, there is provided the system of the first further embodiment or any of the other foregoing embodiments, wherein the external user (**157**) of the ready-to-use medicament is a peritoneal dialysis cyler. According to a fifth further embodiment, there is provided the system of the first further embodiment or any of the other foregoing embodiments, wherein the medicament is removably connected to the fluid circuit by connectors (**124**). According to a sixth further embodiment, there

is provided the system of the first further embodiment or any of the other foregoing embodiments, wherein the pumping and clamping actuators (1016) include a peristaltic pump actuator. According to a seventh further embodiment, there is provided the system of the first further embodiment or any of the other foregoing embodiments, wherein the fluid circuit is connectable to a source of purified water (133). According to an eighth further embodiment, there is provided the system of the first further embodiment or any of the other foregoing embodiments, wherein the fluid circuit is a single-use consumable.

(73) According to a ninth further embodiment, there is provided a system for preparing a medicament for use by a medicament user (157), including: a proportioning machine with a controller (141, 1002) and pumping and clamping actuators (1016) to engage a fluid circuit having pumping and clamping portions that engage with respective actuators among the pumping and clamping actuators (1016); the fluid circuit having a sterilized mixing container (102) connected to the fluid circuit; a concentrate container (199) having concentrated medicament therein; the proportioning machine being configured to flow fluid from the mixing container (102) into and out of the mixing container (102) to circulate the fluid; the proportioning machine being configured to flow water into said mixing container (102) to dilute the concentrated medicament to make a ready-to-use medicament; and the concentrate container (199) being removably connected to the fluid circuit by connectors (124).

(74) According to a tenth further embodiment, there is provided the system of the ninth further embodiment or any of the other foregoing embodiments, wherein the concentrate and ready-to-use medicament are for peritoneal dialysis fluid. According to an eleventh further embodiment, there is provided the system of the ninth further embodiment or any of the other foregoing embodiments, wherein the medicament user (157) of the ready-to-use medicament is a peritoneal dialysis cycler. According to a twelfth further embodiment, there is provided the system of the ninth further embodiment or any of the other foregoing embodiments, wherein the proportioning machine controller (141, 1002) is configured to regulate a clamp (139) on a return line (129) leading to said mixing container (102) to generate a predefined pressure in an outlet line (137, 153) of the fluid circuit which is attachable to an external user (157) of the ready-to-use medicament, wherein the predefined pressure is maintained in the outlet line (137, 153) by pressure feedback control. According to a thirteenth further embodiment, there is provided the system of the twelfth further embodiment or any of the other foregoing embodiments, wherein the clamp (139) is a controllable clamp that regulates flow and pressure in a line (125). According to a fourteenth further embodiment, there is provided the system of the ninth further embodiment or any of the other foregoing embodiments, wherein the pumping and clamping actuators (1016) include a peristaltic pump actuator. According to a fifteenth further embodiment, there is provided the system of the ninth further embodiment or any of the other foregoing embodiments, wherein the fluid circuit is connectable to a source of purified water (133). According to a sixteenth further embodiment, there is provided the system of the ninth further embodiment or any of the other foregoing embodiments, wherein the fluid circuit is a single-use consumable.

(75) According to a seventeenth further embodiment, there is provided a method for preparing a ready-to-use medicament for use by a medicament user (157), including: pumping a first quantity of water into a mixing container (102) in a fluid circuit of a medicament preparation system, the first quantity of water being less than a total quantity of water required in a final batch of medicament; flowing concentrated medicament from a medicament container (199) to the mixing container (102) to form a fluid with the first quantity of water, and pumping the fluid in a circular path through the mixing container (102) to form a first mixed fluid; performing a first conductivity measurement on the first mixed fluid; in response to a controller (141, 1002) of the medicament preparation system determining there is no error in a result of the first conductivity measurement, adding a second quantity of water to the first mixed fluid and pumping the first mixed fluid with the second quantity of water in the circular path through the mixing container (102) to form a

second mixed fluid, the second quantity of water being less than a remaining quantity of water required in the final batch of medicament; performing a second conductivity measurement on the second mixed fluid; and in response to the controller (**141, 1002**) determining there is no error in a result of the second conductivity measurement, adding a third quantity of water to the second mixed fluid and pumping the second mixed fluid with the third quantity of water in the circular path through the mixing container (**102**) to form the final batch of medicament; performing a third conductivity measurement on the final batch of medicament; and in response to the controller (**141, 1002**) determining there is no error in a result of the third conductivity measurement, pumping the final batch of medicament to the medicament user (**157**).

(76) According to an eighteenth further embodiment, there is provided the method of the seventeenth further embodiment or any of the other foregoing embodiments, wherein the mixing container (**102**) is detachably connected to mixing container lines (**123, 125**) of the fluid circuit by connectors (**124**). According to a nineteenth further embodiment, there is provided the method of the seventeenth further embodiment or any of the other foregoing embodiments, wherein the medicament container (**199**) is detachably connected to the fluid circuit by a connector (**124**). According to a twentieth further embodiment, there is provided the method of the seventeenth further embodiment or any of the other foregoing embodiments, wherein the fluid circuit includes a first mixing line (**123**) connected to the mixing container (**102**), a second mixing line (**125**) connected to the mixing container (**102**) and the first mixing line (**123**), a water line (**142**) connecting a water (**133**) source to the first and second mixing lines (**123, 125**), a drain conductivity line (**147**) connected to the first and second mixing lines (**123, 125**), a medicament user supply line (**137, 153**) connecting the medicament user (**157**) to the first and second mixing lines (**123, 125**), and a concentrated medicament supply line (**249**) connecting the medicament container to the first mixing line (**123**), and wherein the pumping of the first quantity of water into the mixing container (**102**) includes pumping the first quantity of water into the mixing container (**102**) through the water line (**142**) and the first mixing line (**123**) at a pressure below a cracking pressure of a first check valve (**154**) on the second mixing line (**125**) while a water line valve (**138**) on the water line (**142**) is open, and a drain conductivity line valve (**140**) on the drain conductivity line (**147**), a batch release valve (**136**) on the medicament user supply line (**137, 153**), and a concentrated medicament supply valve (**144**) on the concentrated medicament supply line (**249**) are closed.

(77) According to a twenty-first further embodiment, there is provided the method of the twentieth further embodiment or any of the other foregoing embodiments, wherein the pumping of the fluid in the circular path through the mixing container (**102**) includes pumping the fluid in the circular path sequentially through the first mixing line (**123**), the second mixing line (**125**), and the mixing container (**102**) while the water line valve (**138**), the drain conductivity line valve (**140**), the batch release valve (**136**), and the concentrated medicament supply valve (**144**) are closed.

(78) According to a twenty-second further embodiment, there is provided the method of the twentieth further embodiment or any of the other foregoing embodiments, wherein the performing of the first conductivity measurement includes pumping a portion of the first mixed fluid through a conductivity sensor (**159c, 159s, 1014**) in the drain conductivity line (**147**) while the drain conductivity line valve (**140**) is open, and the water line valve (**138**), the batch release valve (**136**), and the concentrated medicament supply valve (**144**) are closed.

(79) According to a twenty-third further embodiment, there is provided the method of the twentieth further embodiment or any of the other foregoing embodiments, wherein the pumping of the final batch of medicament to the medicament user (**157**) includes pumping the final batch of medicament through the medicament user supply line (**137, 153**) at a cracking pressure of the first check valve (**154**) while the batch release valve (**136**) is open, and the water line valve (**138**), the drain conductivity line valve (**140**), and the concentrated medicament supply valve (**144**) are closed, and wherein a second check valve (**151**) on the medicament user supply line (**137, 153**) has a cracking

pressure that is lower than the cracking pressure of the first check valve (154).

(80) According to a twenty-fourth further embodiment, there is provided the method of the seventeenth further embodiment or any of the other foregoing embodiments, wherein the fluid circuit includes a first mixing line (123) connected to the mixing container (102), a second mixing line (125) connected to the mixing container (102) and the first mixing line (123), a water line (142) connecting a water source (133) to the first and second mixing lines (123, 125), a drain conductivity line (147) connected to the first and second mixing lines (123, 125), and a medicament user supply line (137, 153) connecting the medicament user (157) to the first and second mixing lines (123, 125), and a concentrated medicament supply line (249) connecting the medicament container to the first mixing line (123), and wherein the pumping of the first quantity of water into the mixing container (102) includes pumping the first quantity of water into the mixing container (102) through the water line (142) and the first mixing line (123) while a water line valve (138) on the water line (142) is open, and a drain conductivity line valve (140) on the drain conductivity line (147), a batch release valve (136) on the medicament user supply line (137, 153), and a concentrated medicament supply valve (144) on the concentrated medicament supply line (249) are closed.

(81) According to a twenty-fifth further embodiment, there is provided the method of the twenty-fourth further embodiment or any of the other foregoing embodiments, wherein a mixing valve (139) leading to the second mixing line (125) is closed during the pumping of the first quantity of water into the mixing container (102). According to a twenty-sixth further embodiment, there is provided the method of the twenty-fourth further embodiment or any of the other foregoing embodiments, wherein a mixing valve (139) leading to the second mixing line (125) is open during the pumping of the first quantity of water into the mixing container (102). According to a twenty-seventh further embodiment, there is provided the method of the twenty-fourth further embodiment or any of the other foregoing embodiments, wherein the performing of the first conductivity measurement includes pumping a portion of the first mixed fluid through a conductivity sensor (159c, 159s, 1014) in the drain conductivity line (137) while the drain conductivity line valve (140) is open, and the water line valve (138), a mixing valve (139) leading to the second mixing line (125), the batch release valve (136), and the concentrated medicament supply valve (144) are closed.

(82) According to a twenty-eighth further embodiment, there is provided the method of the twenty-fourth further embodiment or any of the other foregoing embodiments, wherein the pumping of the final batch of medicament to the medicament user (157) includes pumping the final batch of medicament through the medicament user supply line (137, 153) while the batch release valve (136) is open, and the water line valve (138), a mixing valve (139) leading to the second mixing line (125), the drain conductivity line valve (140), and the concentrated medicament supply valve (144) are closed.

(83) According to a twenty-ninth further embodiment, there is provided the method of the twenty-fourth further embodiment or any of the other foregoing embodiments, wherein the pumping of the final batch of medicament to the medicament user (157) includes pumping the final batch of medicament through the medicament user supply line (137, 153) while the batch release valve (136) and a mixing valve (139) leading to the second mixing line (125) are open, and the water line valve (138), the drain conductivity line valve (140), and the concentrated medicament supply valve (144) are closed.

(84) According to a thirtieth further embodiment, there is provided the method of the seventeenth further embodiment or any of the other foregoing embodiments, wherein the performing of the first conductivity measurement includes: pumping a first quantity of the first mixed fluid through a conductivity sensor (159c, 159s, 1014) and measuring, by the conductivity sensor (159c, 159s, 1014), a conductivity of the first quantity of the first mixed fluid; in response to the controller (141, 1002) determining that a magnitude of the measured conductivity of the first quantity of the first

mixed fluid is not greater than a predefined magnitude, pumping a second quantity of the first mixed fluid through the conductivity sensor (159c, 159s, 1014) and measuring, by the conductivity sensor (159c, 159s, 1014), a conductivity of the second quantity of the first mixed fluid; and in response to the controller (141, 1002) determining that the measured conductivity of the second quantity of the first mixed fluid differs from the measured conductivity of the first quantity of the first mixed fluid by less than a predefined range, outputting, by the controller (141, 1002), a measurement based on either one or both of the measured conductivity of the first quantity of the first mixed fluid and the measured conductivity of the second quantity of the first mixed fluid.

(85) According to a thirty-first further embodiment, there is provided the method of the seventeenth further embodiment or any of the other foregoing embodiments, wherein the performing of the first conductivity measurement includes: pumping a first quantity of the first mixed fluid through a conductivity sensor (159c, 159s, 1014) and measuring, by the conductivity sensor (159c, 159s, 1014), a conductivity of the first quantity of the first mixed fluid; in response to the controller (141, 1002) determining that a magnitude of the measured conductivity of the first quantity of the mixed fluid is not greater than a first predefined magnitude, pumping a second quantity of the first mixed fluid through the conductivity sensor (159c, 159s, 1014) and measuring, by the conductivity sensor (159c, 159s, 1014), a conductivity of the second quantity of the first mixed fluid; in response to the controller (141, 1002) determining that the measured conductivity of the second quantity of the first mixed fluid differs from the measured conductivity of the first quantity of the first mixed fluid by more than a predefined range, further mixing the first mixed fluid through the mixing container (102) and subsequently pumping a third quantity of the first mixed fluid through the conductivity sensor (159c, 159s, 1014) and measuring, by the conductivity sensor (159c, 159s, 1014), a conductivity of the third quantity of the first mixed fluid; in response the controller (141, 1002) determining that a magnitude of the measured conductivity of the third quantity of the further mixed fluid is not greater than a second predefined magnitude, pumping a fourth quantity of the first mixed fluid through the conductivity sensor (159c, 159s, 1014) and measuring, by the conductivity sensor (159c, 159s, 1014), a conductivity of the fourth quantity of the first mixed fluid; and in response to the controller (141, 1002) determining that the measured conductivity of the fourth quantity of the first mixed fluid differs from the measured conductivity of the third quantity of the first mixed fluid by less than a predefined range, outputting, by the controller (141, 1002), a measurement based on either one or both of the measured conductivity of the third quantity of the first mixed fluid and the measured conductivity of the fourth quantity of the first mixed fluid.

(86) It is, thus, apparent that there is provided, in accordance with the present disclosure, Medicament Preparation Devices, Methods, and Systems. Many alternatives, modifications, and variations are enabled by the present disclosure. Features of the disclosed embodiments can be combined, rearranged, omitted, etc., within the scope of the invention to produce additional embodiments. Furthermore, certain features may sometimes be used to advantage without a corresponding use of other features. Accordingly, Applicants intend to embrace all such alternatives, modifications, equivalents, and variations that are within the spirit and scope of the present invention.

Claims

1. A method for preparing a ready-to-use medicament for use by a medicament user, comprising: pumping a first quantity of water into a mixing container in a fluid circuit of a medicament preparation system, the first quantity of water being less than a total quantity of water required in a final batch of medicament; flowing concentrated medicament from a medicament container to the mixing container to form a fluid with the first quantity of water, and pumping the fluid in a circular path through the mixing container to form a first mixed fluid; performing a first conductivity

measurement on the first mixed fluid; in response to a controller of the medicament preparation system determining there is no error in a result of the first conductivity measurement, adding a second quantity of water to the first mixed fluid and pumping the first mixed fluid with the second quantity of water in the circular path through the mixing container to form a second mixed fluid, the second quantity of water being less than a remaining quantity of water required in the final batch of medicament; performing a second conductivity measurement on the second mixed fluid; and in response to the controller determining there is no error in a result of the second conductivity measurement, adding a third quantity of water to the second mixed fluid and pumping the second mixed fluid with the third quantity of water in the circular path through the mixing container to form the final batch of medicament; performing a third conductivity measurement on the final batch of medicament; and in response to the controller determining there is no error in a result of the third conductivity measurement, pumping the final batch of medicament to the medicament user.

2. The method of claim 1, wherein the mixing container is detachably connected to mixing container lines of the fluid circuit by connectors.

3. The method of claim 1, wherein the medicament container is detachably connected to the fluid circuit by a connector.

4. The method of claim 1, wherein the fluid circuit includes a first mixing line connected to the mixing container, a second mixing line connected to the mixing container and the first mixing line, a water line connecting a water source to the first and second mixing lines, a drain conductivity line connected to the first and second mixing lines, a medicament user supply line connecting the medicament user to the first and second mixing lines, and a concentrated medicament supply line connecting the medicament container to the first mixing line, and wherein the pumping of the first quantity of water into the mixing container includes pumping the first quantity of water into the mixing container through the water line and the first mixing line at a pressure below a cracking pressure of a first check valve on the second mixing line while a water line valve on the water line is open, and a drain conductivity line valve on the drain conductivity line, a batch release valve on the medicament user supply line, and a concentrated medicament supply valve on the concentrated medicament supply line are closed.

5. The method of claim 4, wherein the pumping of the fluid in the circular path through the mixing container includes pumping the fluid in the circular path sequentially through the first mixing line, the second mixing line, and the mixing container while the water line valve, the drain conductivity line valve, the batch release valve, and the concentrated medicament supply valve are closed.

6. The method of claim 4, wherein the performing of the first conductivity measurement includes pumping a portion of the first mixed fluid through a conductivity sensor in the drain conductivity line while the drain conductivity line valve is open, and the water line valve, the batch release valve, and the concentrated medicament supply valve are closed.

7. The method of claim 4, wherein the pumping of the final batch of medicament to the medicament user includes pumping the final batch of medicament through the medicament user supply line at a cracking pressure of the first check valve while the batch release valve is open, and the water line valve, the drain conductivity line valve, and the concentrated medicament supply valve are closed, and wherein a second check valve on the medicament user supply line has a cracking pressure that is lower than the cracking pressure of the first check valve.

8. The method of claim 1, wherein the fluid circuit includes a first mixing line connected to the mixing container, a second mixing line connected to the mixing container and the first mixing line, a water line connecting a water source to the first and second mixing lines, a drain conductivity line connected to the first and second mixing lines, a medicament user supply line connecting the medicament user to the first and second mixing lines, and a concentrated medicament supply line connecting the medicament container to the first mixing line, and wherein the pumping of the first quantity of water into the mixing container includes pumping the first quantity of water into the

mixing container through the water line and the first mixing line while a water line valve on the water line is open, and a drain conductivity line valve on the drain conductivity line, a batch release valve on the medicament user supply line, and a concentrated medicament supply valve on the concentrated medicament supply line are closed.

9. The method of claim 1, wherein the performing of the first conductivity measurement includes: pumping a first quantity of the first mixed fluid through a conductivity sensor and measuring, by the conductivity sensor, a conductivity of the first quantity of the first mixed fluid; in response to the controller determining that a magnitude of the measured conductivity of the first quantity of the first mixed fluid is not greater than a predefined magnitude, pumping a second quantity of the first mixed fluid through the conductivity sensor and measuring, by the conductivity sensor, a conductivity of the second quantity of the first mixed fluid; and in response to the controller determining that the measured conductivity of the second quantity of the first mixed fluid differs from the measured conductivity of the first quantity of the first mixed fluid by less than a predefined range, outputting, by the controller, a measurement based on either one or both of the measured conductivity of the first quantity of the first mixed fluid and the measured conductivity of the second quantity of the first mixed fluid.

10. The method of claim 1, wherein the performing of the first conductivity measurement includes: pumping a first quantity of the first mixed fluid through a conductivity sensor and measuring, by the conductivity sensor, a conductivity of the first quantity of the first mixed fluid; in response to the controller determining that a magnitude of the measured conductivity of the first quantity of the first mixed fluid is not greater than a first predefined magnitude, pumping a second quantity of the first mixed fluid through the conductivity sensor and measuring, by the conductivity sensor, a conductivity of the second quantity of the first mixed fluid; in response to the controller determining that the measured conductivity of the second quantity of the first mixed fluid differs from the measured conductivity of the first quantity of the first mixed fluid by more than a predefined range, further mixing the first mixed fluid through the mixing container and subsequently pumping a third quantity of the first mixed fluid through the conductivity sensor and measuring, by the conductivity sensor, a conductivity of the third quantity of the first mixed fluid; in response the controller determining that a magnitude of the measured conductivity of the third quantity of the further mixed fluid is not greater than a second predefined magnitude, pumping a fourth quantity of the first mixed fluid through the conductivity sensor and measuring, by the conductivity sensor, a conductivity of the fourth quantity of the first mixed fluid; and in response to the controller determining that the measured conductivity of the fourth quantity of the first mixed fluid differs from the measured conductivity of the third quantity of the first mixed fluid by less than a predefined range, outputting, by the controller, a measurement based on either one or both of the measured conductivity of the third quantity of the first mixed fluid and the measured conductivity of the fourth quantity of the first mixed fluid.
